1 Important Inequalities

The proofs of the error bound formulas all use the *Triangle Inequality*

$$\left|\sum_{i=1}^{n} x_i\right| \leq \sum_{i=1}^{n} |x_i|,$$

which you proved in AEP #1, as well as the following similar-looking inequality for integrals.

Proposition 1.1 (Triangle inequality for integrals)**.** *If $f : [a,b] \to \mathbb{R}$ is integrable then so is $|f|$ and*

$$\left|\int_a^b f(x)dx\right| \leq \int_a^b |f(x)|dx.$$

1. Assume the first statement in Proposition 1.1 and prove the second.
(*Hint*: $-|f(x)| \leq f(x) \leq |f(x)|$.)

2 Left-Endpoint Rule

Theorem 2.1 (Error bound for left- and right-endpoint approximations)**.** *Let $E_n^L(f)$ and $E_n^R(f)$ denote the absolute error $\left|\sum_{i=1}^n f(x_i^*)\Delta x - \int_a^b f(x)dx\right|$ using the left- and right-endpoint approximations, respectively. If $M = \max\{|f'(x)|\}_{x\in[a,b]}$, then E_n^L and E_n^R are both at most $M\frac{(b-a)^2}{2n}$.*

Proof. (a) For the left-endpoint approximation, show that the absolute error $E_n^L(f)$ can be written as

$$E_n^L(f) = \left|\sum_{i=1}^n L_i\right|,$$

where

$$L_i := f(x_{i-1})\left(\frac{b-a}{n}\right) - \int_{x_{i-1}}^{x_i} f(x)dx. \tag{2.2}$$

(*Hint*: Start with the definition $E_n^L(f) = \left|\sum_{i=1}^n f(x_i^*)\Delta x - \int_a^b f(x)dx\right|$ and write $\int_a^b f(x)dx = \sum_{i=1}^n \int_{x_{i-1}}^{x_i} f(x)dx$.)

(b) Show that Equation (2.2) can be written as

$$L_i = \int_{x_{i-1}}^{x_i} (t - x_i) f'(t) dt. \tag{2.3}$$

(*Hint*: Integrate Equation (2.3) by parts.)

(c) Assume now that $M = \max\{|f'(x)|\}_{x\in[a,b]}$. Substitute Equation (2.3) into $E_n^L(f) = |\sum_{i=1}^n L_i|$ and use the Triangle Inequality together with Proposition 1.1 to show that

$$E_n^L(f) \leq M\frac{(b-a)^2}{2n}.$$

□